

Question: 1

Create a side-by-side comparison table showing at least 4 key differences between the OSI Model and the TCP/IP Model, including the number of layers, layer names, and where protocols like HTTP, TCP, and IP fit?

Answer: Comparison Table

Feature	OSI Model	TCP/IP Model
Number of Layers	7 Layers	4 Layers (or 5 in modern updates)
Layer Names	Application, Presentation, Session, Transport, Network, Data Link, Physical	Application, Transport, Internet, Network Access
Application Protocol (HTTP)	Fits strictly into Layer 7 (Application)	Fits into the Application Layer (combines OSI layers 5, 6, & 7)
Transport Protocol (TCP)	Fits strictly into Layer 4 (Transport)	Fits into the Transport Layer (Host-to-Host)
Internet Protocol (IP)	Fits strictly into Layer 3 (Network)	Fits into the Internet Layer
Development Approach	Developed as a theoretical, generic model before protocols were invented.	Developed as a practical framework based around existing protocols.

Question: 2

Open Wireshark, start a capture on your active network interface, and visit any website (like www.youtube.com) in your browser. Stop the capture and find a TCP packet in the list. Write down the source IP, destination IP, source port, destination port, and the TCP flag(s) set for that packet?

Answer: 1. How to Find the Packet

Filter: Type tcp in the top bar and press Enter.

Selection: Click on any row that shows TCP in the Protocol column.

2. Packet Details to Write Down

Look at the Internet Protocol (IPv4) and Transmission Control Protocol (TCP) sections in the middle pane to record these values:

Source IP: [Copy the IP address under Source in the packet list, e.g., 192.168.1.15]

Destination IP: [Copy the IP address under Destination, e.g., 142.250.190.46]

Source Port: [Expand the TCP section to find the Source Port, e.g., 53245]

Destination Port: [Find the Destination Port in the TCP section, e.g., 443]

TCP Flags Set: [Expand the Flags submenu inside the TCP section. Note down which flag is active, e.g., [SYN] or [ACK]]

Question: 3

In Wireshark, filter packets by 'udp' and find a DNS query. Identify and note which TCP/IP layer each of the following fields belongs to: source IP, destination IP, source port, destination port, DNS query name?

Answer: Field to Layer Mapping

Source IP

→right arrow

→

Internet Layer (Handles data routing across networks).

Destination IP

→right arrow

→

Internet Layer (Handles data routing across networks).

Source Port

→right arrow

→

Transport Layer (Manages host-to-host process communication).

Destination Port

→right arrow

→

Transport Layer (Manages host-to-host process communication; DNS uses port 53).

DNS Query Name

→right arrow

→

Application Layer (Handles user-facing network service data, such as requesting a domain name like example.com)

Question: 4

Analyze any captured IPv6 packet in Wireshark and list the differences you observe in the IP header compared to an IPv4 packet.

Hint: Look for differences in address format, header length, and any extra fields present in IPv6.

Answer: Core Header Differences

Address Format & Size: IPv6 uses 128-bit hexadecimal addresses (e.g., fe80::1), whereas IPv4 uses 32-bit dotted-decimal addresses (e.g., 192.168.1.1).

Header Length: IPv6 has a fixed header length of 40 bytes, while IPv4 has a variable header length (minimum 20 bytes) depending on optional fields.

Simplified Fields: IPv6 removes fields like IHL (Internet Header Length), Identification, Flags, and Fragment Offset from the base header to speed up processing.

Time-to-Live (TTL) Rename: The TTL field in IPv4 is renamed to Hop Limit in IPv6.

Protocol Field Rename: The Protocol field in IPv4 is renamed to Next Header in IPv6.

New Traffic Field: IPv6 introduces a Flow Label field (20-bit) to maintain the sequential flow of packets belonging to a specific communication stream.